

MICHIGAN CITY MUNICIPAL, IN, USA

WMO: 726358

Lat: 41.703N

Lon: 86.821W

Elev: 657

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 00384

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.5	4.8	-9.3	3.4	1.8	-4.2	4.5	8.1	24.5	26.0	22.0	22.9	9.0	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	88.5	72.8	86.1	71.5	82.3	69.6	76.0	84.2	74.3	81.9	72.9	79.9	8.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	126.5	77.9	72.3	122.6	77.4	71.5	119.3	76.9	39.8	84.3	38.3	82.8	36.9	81.1	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.0	15.9	-5.8	91.6	10.6	3.9	-13.4	94.4	-19.7	96.7	-25.6	98.9	-33.3	101.7		
(5)			-6.5	78.6	10.5	2.2	-14.1	80.2	-20.3	81.5	-26.2	82.7	-33.8	84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	24.6	28.1	37.1	47.2	58.4	66.5	70.7	69.2	64.7	52.8	39.8	32.7		
	DBStd	18.37	11.76	11.35	10.85	9.07	9.94	7.55	6.25	5.14	7.94	8.74	9.05	9.62		
	HDD50	2957	786	614	426	157	31	0	0	0	1	72	328	542		
	HDD65	6311	1251	1032	867	538	258	73	19	20	105	390	757	1001		
	CDD50	2755	0	2	27	74	292	496	642	594	442	159	21	6		
	CDD65	631	0	0	3	4	54	119	196	148	95	12	0	0		
	CDH74	5100	0	0	22	41	499	1059	1657	1075	672	74	1	0		
	CDH80	1401	0	0	1	5	117	305	526	246	191	10	0	0		
(14) Wind	WSAvg	6.4	8.5	7.9	7.4	7.4	6.0	5.1	4.5	3.6	4.1	6.7	8.0	7.5		
(15) Precipitation	PrecAvg	39.3	2.6	2.1	2.5	3.5	3.7	4.3	4.1	4.0	3.4	3.5	3.1	2.6		
	PrecMax	56.8	5.7	4.2	5.2	6.8	6.9	10.0	7.2	9.8	13.6	8.9	7.5	5.0		
	PrecMin	29.9	0.8	0.8	0.8	1.1	0.6	1.5	1.6	0.9	0.7	0.8	0.6	0.8		
	PrecStd	6.1	1.3	1.0	1.1	1.5	1.9	2.1	1.5	2.3	2.6	2.0	1.5	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	62.7	79.0	79.2	88.1	90.2	96.6	90.0	90.0	81.2	70.0	61.2		
		MCWB	53.5	54.2	62.7	60.8	67.9	71.0	73.4	73.5	72.7	69.8	59.0	57.9		
	2%	DB	49.7	54.7	68.0	73.0	82.3	86.5	89.6	86.1	86.2	75.1	63.2	56.6		
		MCWB	47.7	50.2	59.1	57.7	65.7	69.3	74.6	72.1	73.9	64.2	56.2	53.4		
	5%	DB	43.0	48.3	61.0	69.5	80.5	82.5	85.9	81.9	81.3	71.1	58.8	50.2		
		MCWB	40.7	45.0	53.6	57.2	64.7	68.0	72.6	70.5	70.9	63.5	50.6	46.5		
	10%	DB	38.3	42.9	54.3	63.6	75.9	80.9	81.7	80.6	77.4	67.7	53.8	45.3		
		MCWB	36.2	38.6	47.8	52.6	63.0	66.9	70.2	69.7	68.5	61.7	48.5	42.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.9	57.2	63.9	63.4	71.2	75.2	78.9	77.3	78.3	71.6	61.3	59.7		
		MCDB	56.4	60.8	76.8	73.7	82.3	86.6	87.3	83.2	83.9	76.9	65.7	61.0		
	2%	WB	48.4	51.5	60.5	60.5	68.9	73.3	76.1	74.9	74.8	67.7	57.2	53.7		
		MCDB	49.6	55.0	68.7	68.2	78.4	81.5	85.6	81.4	82.0	73.2	63.5	55.7		
	5%	WB	41.5	45.0	55.1	57.8	67.2	71.7	74.5	73.1	72.8	64.8	52.1	46.3		
		MCDB	44.1	48.4	58.4	66.1	76.3	80.1	82.7	79.3	79.4	68.8	56.2	48.0		
	10%	WB	37.1	38.9	48.7	53.9	65.3	69.5	73.0	71.8	70.8	62.5	48.3	42.6		
		MCDB	38.6	42.6	54.1	63.3	73.9	77.2	80.1	77.7	75.6	66.1	53.2	45.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.3	15.1	18.6	19.8	19.9	18.4	18.9	19.1	16.4	12.9	10.9		
		MCDBR	15.3	20.7	23.5	26.6	24.2	22.7	21.8	21.6	21.0	20.6	19.3	16.7		
		MCWBR	14.0	17.5	15.6	15.8	11.9	10.8	9.4	10.7	10.3	13.2	12.8	13.9		
	5% WB	MCDBR	15.8	20.5	21.2	23.5	20.8	19.7	19.5	17.3	18.1	17.9	18.0	15.0		
		MCWBR	14.8	17.7	15.5	16.9	11.2	11.2	10.0	10.3	10.5	13.7	13.9	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.328	0.346	0.400	0.430	0.435	0.449	0.446	0.420	0.358	0.323	0.304		
	taud		2.426	2.348	2.374	2.251	2.228	2.263	2.221	2.227	2.261	2.438	2.488	2.485		
	Ebn at Noon		267	277	285	276	269	267	262	259	257	264	259	259		
	Edh at Noon		25	31	34	42	44	43	44	43	38	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		477	741	1131	1476	1683	1928	1945	1687	1388	904	561	400		
	RadStd		47	96	89	125	150	121	97	99	115	96	74	43		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page